

A graphon class without a finite-step representative

Counterexample to Conjecture 00000004408

Abstract

The symmetric graphon $W(x, y) = xy$ on the unit square has a nonatomic distribution of function values. This distribution is unchanged by measure-preserving relabellings and modification on null sets. Consequently, no representative of its equivalence class is essentially countably valued, and in particular no representative is a finite-step graphon. This refutes the first universal assertion of Conjecture 00000004408. The accompanying Lean 4 proof verifies the measure-theoretic obstruction, its invariance under the generated equivalence relation, and the finite-step consequence.

1 Statement and conventions

The source defines L^p reconstruction equivalence as equivalence of graphons under measure-preserving transformations and zero-measure modification. It then asserts that every class has a step representative of minimal step count, that this count is a class invariant, and that for purely jumping graphons it is the number of jumps.

The English and Chinese statements agree on these points. Neither states a restriction to graphons that are already step functions. We use the usual meaning of a finite-step graphon: there is a finite measurable partition $[0, 1] = A_1 \sqcup \cdots \sqcup A_n$, up to null sets, and constants c_{ij} such that $U(x, y) = c_{ij}$ for almost every $(x, y) \in A_i \times A_j$. Refining a partition is permitted. Our obstruction also rules out a countable partition with one constant on each block rectangle. An uncountable partition into singletons is not a finite- or countable-step representation.

Write λ for Lebesgue probability measure on $[0, 1]$ and $m = \lambda \otimes \lambda$. For a measurable, measure-preserving map $\phi : [0, 1] \rightarrow [0, 1]$, let

$$U^\phi(x, y) = U(\phi(x), \phi(y)).$$

We allow these relabellings without requiring invertibility, and take the equivalence relation generated by such relabellings and equality m -almost everywhere. Thus the argument covers, in particular, the convention using only measure-preserving bijections. No limit of relabellings or unspecified topological closure is used. The source's equivalence definition does not depend on a numerical choice of p ; approximation in an L^p norm is a different assertion.

2 The counterexample

Set

$$W : [0, 1]^2 \longrightarrow [0, 1], \quad W(x, y) = xy.$$

This is a graphon: it is continuous, symmetric, and has values between 0 and 1 on its stated domain.

Lemma 1. *For every countable set $S \subseteq \mathbb{R}$,*

$$m\{(x, y) : W(x, y) \in S\} = 0.$$

Proof. For each fixed $x \neq 0$, the map $y \mapsto xy$ is injective, so the set $\{y \in [0, 1] : xy \in S\}$ is countable and has Lebesgue measure zero. The only exceptional value $x = 0$ is itself a null set. The inverse image $W^{-1}(S)$ is measurable because S is countable and W is continuous. Tonelli's theorem, applied to its indicator function, therefore gives

$$m(W^{-1}(S)) = \int_{[0,1]} \lambda\{y \in [0, 1] : xy \in S\} d\lambda(x) = 0.$$

The possible nonzero inner measure at $x = 0$ has no effect on the integral. $\square$

For any graphon U , define its value distribution by the pushforward probability measure

$$\nu_U = U_*m, \quad \nu_U(B) = m(U^{-1}(B)) \quad \text{for every Borel set } B \subseteq \mathbb{R}.$$

Lemma 2. *The measure ν_U is invariant under the reconstruction equivalence defined above.*

Proof. If $U = V$ almost everywhere, their inverse images of every Borel set agree up to a null set, hence $\nu_U = \nu_V$. If $V = U^\phi$, the map $\phi \times \phi$ preserves the product measure m . Indeed, its pushforward agrees with m on measurable rectangles because

$$m((\phi \times \phi)^{-1}(A \times B)) = \lambda(\phi^{-1}(A))\lambda(\phi^{-1}(B)) = \lambda(A)\lambda(B),$$

and these rectangles determine the product probability measure. It follows that

$$\nu_{U^\phi} = U_*((\phi \times \phi)_*m) = U_*m = \nu_U.$$

Equality of measures is reflexive, symmetric, and transitive; invariance therefore passes to the full equivalence relation generated by the two operations, including operations traversed in reverse. $\square$

Theorem 1. *No graphon equivalent to $W(x, y) = xy$ is essentially countably valued. In particular, the equivalence class of W has no finite-step representative.*

Proof. Suppose V is equivalent to W and $V(x, y) \in S$ almost everywhere for a countable set S . Then $\nu_V(S) = 1$. Lemmas 1 and 2 give

$$1 = \nu_V(S) = \nu_W(S) = 0,$$

a contradiction. A finite-step graphon with n blocks is almost everywhere valued in the finite set $\{c_{ij} : 1 \leq i, j \leq n\}$, so it is excluded. The same reasoning applies to countably many blocks, since a countable union of countable sets is countable. $\square$

Corollary 1. *The assertion that every reconstruction-equivalence class contains a finite-step representative, hence one with a minimal finite step count, is false. Therefore Conjecture 00000004408, as stated, is false.*

This conclusion does not require deciding a convention for “number of jumps” in two variables, or separately refuting an assertion limited to graphons already known to have a step representation. It also does not deny that general graphons can be approximated by step functions.

3 Lean verification and the translation to graphons

The project pins Lean v4.19.0 and the following Mathlib commit:
c44e0c8ee63ca166450922a373c7409c5d26b00b.

The proof is in `lean/Main.lean`. The following mathematical steps are checked by the Lean kernel:

1. `productKernel_range`

Together with the measurability and symmetry lemmas, this verifies the graphon conditions for xy .

2. `productKernel_avoids_countable`

This proves Lemma 1 for every countable subset of $\mathbb{R}$, using product measure and nonatomic Lebesgue measure.

3. `equivalence_preserves_valueLaw`

This proves Lemma 2, including its passage to `Relation.EqvGen`.

4. `no_finite_step_in_productKernel_class`

This proves the final nonexistence statement using the more general lemma excluding all essentially countably valued representatives.

For the measure-space model, Lean uses $\mathbb{R}$ with the measure $\lambda_0 = \text{volume}|_{[0,1]}$, then $\lambda_0 \otimes \lambda_0$. The file verifies that both are probability measures. This is the unit-interval model with a null exterior, not a claim that xy has values in $[0, 1]$ on all of $\mathbb{R}^2$. Kernels are real-valued functions that are measurable almost everywhere for this product measure, so null modifications are permitted.

The formal finite-step predicate allows an arbitrary map $b : \mathbb{R} \rightarrow \text{Fin}(n)$ and arbitrary weights c_{ij} , with $U(x, y) = c_{b(x)b(y)}$ almost everywhere. Every usual finite measurable partition gives this form by choosing its block index; null exceptional points may be assigned to any block. The Lean predicate even omits the measurability requirement on b , so its negation excludes a broader class than the finite-step graphons in the source. The passage from a partition written in ordinary graphon language to this index map, and the identification of $[0, 1]$ with its restricted-measure model, are semantic translations explained here; no theorem concluding nonexistence is taken as a Lean assumption.

The source’s informal “step count” and “jump count” are not assigned new definitions. The refuted existence assertion only needs the necessary finite-range property proved in the file. The formal proofs contain no unfinished terms or additional axioms: the printed dependency audit lists only `propext`, `Classical.choice`, and `Quot.sound`. No finite experiment or numerical approximation is used as evidence for the infinite statement.

Source. The Last Math Competition, Conjecture 00000004408. The exact bilingual source and build instructions are included in `SOURCE.md` and `README.md`; snapshot identifiers are in `verification/SOURCE_PROVENANCE.md`.